

VDT — Verifiable Digital Transformation

A proof layer for organizational learning systems

Keywords: Proof of Human Oversight · Digital Transformation · EU AI Act Art.14 · Organizational Learning · KMU Onboarding

Abstract

This concept note proposes a joint research initiative to address the **Proof Gap** in digital transformation: the structural inability of current systems to verify that transformation processes were actually executed as designed. We introduce the PHO Framework (Proof of Human Oversight) as a cryptographic proof layer for organizational learning, with application to the LeBi Interreg project's KMU onboarding methodology.

1. Research Problem

Digital transformation frameworks — in learning, onboarding, and organizational change — operate under a structural blind spot: **they can document process design, but not process proof.**

Current systems measure adoption perception, training completion rates, and self-reported compliance. What they cannot measure is: who made which decision, when, under which authorization, and whether the agreed process was actually followed — not just recorded.

We define this as the **Proof Gap**: the distance between documented transformation and verifiable transformation.

2. Research Question

*"How can digital transformation and organizational learning processes be made **independently auditable and cryptographically verifiable** — not only documented — using proof-of-oversight mechanisms?"*

Sub-questions: (a) What constitutes valid proof of human decision in a learning process? (b) How can PHO be operationalized in onboarding? (c) What is the minimum viable proof layer for EU AI Act Article 14 compliance?

3. Methodology

Empirical base: 16+ KMU pilot companies from the LeBi Interreg project (Vorarlberg, Ostschweiz, Süddeutschland), Q1 2026 – Q4 2028.

Intervention: Integration of a PHO layer into LeBi's onboarding toolbox. Each critical decision step is anchored to an immutable ledger record: actor DID (decentralized identifier), timestamp, authorization level, process deviation (if any).

Measurement: Comparative analysis of (a) documented process vs (b) proven process execution. Gap quantification. Human oversight coverage score (SGE).

4. Expected Outputs

Academic:

- Peer-reviewed paper: "From Digital Transformation to Verifiable Transformation"
- Framework definition and validation methodology
- Empirical dataset from 16+ KMU pilots

Practical:

- Replicable PHO-onboarding integration model
- Open methodology for KMU adoption
- Certification pathway aligned with EU AI Act Art. 14

Societal:

- First empirical evidence base for proof-of-oversight in organizational learning in German-speaking Europe

5. EU Regulatory Alignment

This initiative directly addresses **EU AI Act Article 14** (Human Oversight) and **GDPR Article 22** requirements for meaningful human involvement in automated systems. The PHO Framework was designed as a constitutional primitive, not a compliance patch.

The GRC (Governance, Risk, Compliance) market is projected at €65B globally. A proof-based verification layer positions this research at the intersection of academic rigor and market relevance.

6. Partnership Model

IDT Kempton brings	PHO Framework brings
Academic credibility	Proof of Human Oversight (PHO)
Access to real KMUs	Decentralized Identity (DID)
LeBi project infrastructure	Immutable ledger technology
Interreg / EU network	Production-ready system
Institutional language	EU AI Act alignment

This is an initial working hypothesis intended to be refined together with IDT's research perspective. The framework presented here is functional but open — the academic framing, research design, and pilot scope are all areas where IDT's expertise would be essential to shape the final direction.